Sameer Choudhary

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.25
Jawahar Navodaya Vidyalaya Meja Prayagraj UP (74001)	Class XII (CBSE)	2023	85.20%
Jawahar Navodaya Vidyalaya Nud Samba Jammu And Kashmir (24908)	Class X (CBSE)	2021	77.40%

EXPERIENCE

Research Intern - University of Kashmir

- Proposed and implemented a **novel analytical model** for the **Third-Order Correlation Function (3PCF)** , advancing higher-order statistical modeling of large-scale cosmological structures.
- Built end-to-end data processing and statistical validation pipelines for **millions of observational and simulation records,** parameter estimation, covariance-based analysis, residual diagnostics, and **^{2} goodness-of-fit** testing.
- Engineered reproducible scientific computing workflows using **Python, NumPy, and SciPy** to automate model evaluation and benchmark theoretical predictions across **1-50 Mpc** spatial scales.

PROJECTS

LakeFlow - Full-Stack Data Engineering Platform

- Developed a full-stack data engineering platform with automated ETL workflows, enabling dataset upload, processing, and analysis using Python (Pandas, NumPy), Flask, and MySQL.
- Built 20+ REST APIs, designed a 10+ table normalized MySQL schema, and implemented JWT authentication with OOP-based modular architecture for scalable pipeline management.

Book Recommendation System

- Built an item-based collaborative filtering (KNN) recommendation system on a dataset of 1.1M+ ratings, 278K+ users, and 271K+ books, generating personalized Top-5 recommendations.
- Optimized the recommendation pipeline using SciPy CSR sparse matrices, enabling efficient similarity computation and fast nearest-neighbor search on large-scale data.

Bayesian Signal Identification from Noisy Time-Series Data

- Developed a Python-based Bayesian inference pipeline using Metropolis-Hastings MCMC to recover gravitational-wave burst signals from a 2,000-point simulated time-series dataset.
- Implemented waveform modeling, log-likelihood (^{2}) evaluation, posterior sampling, and estimation of 3 physical parameters, reconstructing best-fit waveforms through statistical analysis and visualization.

Time-Series Periodicity Detection using Phase Distance Correlation (Github)

- Worked on framework for detecting hidden periodicities in long-duration time-series data.
- Implemented Phase Distance Correlation techniques for feature extraction.

X-ray Timing and Quasi-Periodic Oscillation Analysis (Github)

- Performed spectral timing analysis of astrophysical X-ray light curves.
- Applied Fourier-based power spectral density estimation techniques and identified quasi-periodic oscillation signatures.

SKILLS

Programming Languages : Python, C++, C, JavaScript, MySQL.

Machine Learning & Data Science: Scikit-learn, NumPy, Pandas, SciPy, Matplotlib, Machine Learning, Bayesian Inference, MCMC Sampling, Recommendation Systems, Clustering, Regression, Statistical Modelling, Data Preprocessing.

Tools & Core Concepts: Git, GitHub, Linux, macOS, Jupyter Notebook, Google Colab, VS Code, LaTeX, Data Structures & Algorithms, Signal Processing.

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SCHOLASTIC ACHIEVEMENTS

Publication - JCAP Submission (link):

- Proposed and implemented a novel analytical model for the **Third-Order Correlation Function (3PCF),** advancing higher-order statistical modeling. Engineered data processing and validation pipelines for large observational datasets.
- Submitted the work to **Top Teir-1 Journal** (JCAP), 2025.

POSITIONS OF RESPONSIBILITY

Overall Head, Cepheid (Astronomy Club), IIT Hyderabad

- Lead club operations, organize technical seminars, coordinated outreach engaging 500+ students at National Space Day. Forged collaborations with faculty, student organizations, and national & international experts.

Internship Coordinator, Office of Career Services, IIT Hyderabad

- Assist in institute-level internship operations, Facilitate communication between recruiters and students.

Coordinator, Finance and Consulting Club (FCC), IIT Hyderabad

- Organized finance-related learning sessions and discussions, organised participation in finance and consulting competitions.

Coordinator, Torque Club, IIT Hyderabad

- Coordinated technical and organizational activities of the institute automotive club, with student teams to execute club initiatives.

RELEVANT COURSES

Relevant Coursework : Introduction to Artificial Intelligence, Data Structures and Algorithms, Probability and Statistics, Linear Algebra, Computational Physics, Differential Equations, Mathematical Physics, Introduction to Entrepreneurship.

EXTRACURRICULAR

Astronomy Outreach Activities

- Organized telescope observation sessions, public programs and astronomy awareness activities through Clubs.

NSO Volleyball Team Member, IIT Hyderabad

- Represented IIT Hyderabad in volleyball under the National Sports Organization program, part of Institute League.